VANISHING OF HOM AND EXT FROM A PERFECT COMPLEX INTO COMPLEXES WITH A COHOMOLOGICAL GAP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Maps from a perfect complex into any complex are computed as the degree zero cohomology of the tensor with the dual perfect complex. invoked at proof lines 3483-3488. Converts the Hom vanishing into a cohomology vanishing for a tensor product.
- The tensor product of the dual with L has cohomology controlled by the spectral sequence from the tor amplitude of the dual and the cohomology sheaves of L. used at proof lines 3494-3500. Shows the cohomology of the tensor vanishes in a range when L has a cohomological gap.
- On a quasi-compact quasi-separated scheme the degree zero cohomology depends only on a finite stretch of cohomology sheaves, controlled by the number of affines needed for a cover. invoked at proof lines 3502-3514. Lets the vanishing of a finite range of cohomology sheaves force the vanishing of H^0.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 3483-3488 for the Hom rewriting; the tor amplitude choice and spectral sequence are inside the counted span.

Source reference. Original lemma 08DX: sources/stacks-perfect/pinned-original-source.tex, source lines 620–620. Cited at proof lines 3489-3492 for the tensor being in D_QCoh; the spectral sequence argument is inside the counted span.

Source reference. Original lemma 08E5: sources/stacks-perfect/pinned-original-source.tex, source lines 2422–2422. Cited at proof lines 3489-3492 to see the perfect dual has quasi-coherent cohomology sheaves; the use in the spectral sequence is inside the counted span.

Source reference. Original lemma 071L: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 3502-3510 to fix N and reduce H^0 to a finite stretch; the truncation and vanishing argument is inside the counted span.

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Source reference. Original lemma 08D4: sources/stacks-perfect/pinned-original-source.tex, source lines 375–375. Cited at proof lines 3510-3512 to identify H^0 with that of the truncation; the final vanishing conclusion is inside the counted span.

Source reference. Original lemma 05TC: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 3516-3521 for the final vanishing; the assembly of the interval [-b - N, -a] is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let K be a perfect object of $D(\mathcal{O}_X)$. Then*

1. *there exist integers $a \leq b$ such that for any $L \in D_{QCoh}(\mathcal{O}_X)$ with $H^i(L) = 0$ for $i \in [a, b]$ we have $\mathrm{Hom}_{D(\mathcal{O}_X)}(K, L) = 0$, and*
2. *if L is bounded, then $\mathrm{Ext}^n_{D(\mathcal{O}_X)}(K, L)$ is zero for all but finitely many n.*

Proof. Part (2) follows from (1) as $\mathrm{Ext}^n_{D(\mathcal{O}_X)}(K, L) = \mathrm{Hom}_{D(\mathcal{O}_X)}(K, L[n])$. We prove (1). Since K is perfect we have

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K, L) = H^0(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L)$$

where $K^\vee$ is the "dual" perfect complex to K, see Cohomology, Lemma 08DQ. Note that $K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ is in $D_{QCoh}(X)$ by Lemmas 08DX and 08E5 (to see that a perfect complex has quasi-coherent cohomology sheaves). Say $K^\vee$ has tor amplitude in $[a, b]$. Then the spectral sequence

$$E_1^{p,q} = H^p(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} H^q(L)) \Rightarrow H^{p+q}(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L)$$

shows that $H^j(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L)$ is zero if $H^q(L) = 0$ for $q \in [j - b, j - a]$. Let N be the integer d of Cohomology of Schemes, Lemma 071L. Then $H^0(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L)$ vanishes if the cohomology sheaves

$$H^{-N}(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L),\ H^{-N+1}(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L),\ \ldots,\ H^0(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L)$$

are zero. Namely, by the lemma cited and Lemma 08D4, we have

$$H^0(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L) = H^0(X, \tau_{\geq -N}(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L))$$

and by the vanishing of cohomology sheaves, this is equal to $H^0(X, \tau_{\geq 1}(K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} L))$ which is zero by Derived Categories, Lemma 05TC. It follows that $\mathrm{Hom}_{D(\mathcal{O}_X)}(K, L)$ is zero if $H^i(L) = 0$ for $i \in [-b - N, -a]$. □